

Fridge TEMPERATURE AND HUMIDITY Indicator

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We can measure temperature and humidity inside the fridge using a normal temperature-humidity indicator but relative humidity (RH) could be inaccurate in that case. The moment the fridge door is opened, RH will shoot up due to ingress or egress of moisture in the surroundings.

The small sniffer device, described in this article, picks up temperature and humidity from inside the fridge and transmits on an RF link to a nearby receiver unit. The receiver unit checks the received code, identifies the right sniffer device and displays live temperature and humidity. Author's prototype is shown in Fig. 1.



Fig. 1: Author's prototype

PARTS LIST

Semiconductors:

IC1, IC2	- ATmega328P MCU with Arduino Uno bootloader
IC3	- 7805 voltage regulator
TX1	- 433MHz RF transmitter module
RX1	- 433MHz RF receiver module
ZD1	- 5V zener diode
ZD2	- 3.3V zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2, R5 - 10-kilo-ohm
R3 - 330-ohm
R4 - 100-ohm
VR1 - 10-kilo-ohm preset

Capacitors

C1, C2, C9, C10 - 22pF ceramic disk
C3, C4 - 100μF, 25V electrolytic
C5, C6, C11 - 100nF ceramic disk
C7 - 10μF, 16V electrolytic
C8 - 470μF, 25V electrolytic

Miscellaneous:

BATT.1, BATT.2	- 9V battery
ANT1, ANT2	- Spiral antenna
XTAL1, XTAL2	- 16MHz crystal oscillator
LCD1	- 16x2 character module display
SI, S2	- Tactile switch
SENSOR1	- AM2302 digital temperature and humidity sensor
CON1	- 4-pin connector for sensor

Circuit and working

Circuit diagram of the transmitter unit is shown in Fig. 2. It is built around ATmega328P microcontroller (MCU) (IC1) with Arduino Uno bootloader, AM2302 digital temperature and humidity sensor connected as SENSOR1 to CON1, 433MHz transmitter (TX1), 5V zener ZD1, 3.3V zener ZD2 and a few other components.

Low power consumption of the transmitter is the essence for long operating hours of the gadget. For that, the conventional regulator is replaced by 5V and 3.3V zener diodes, with 330-ohm resistor R3 and 100-ohm resistor R4 in series to reduce current consumption.

Circuit diagram of the receiver unit is shown in Fig. 3. It is built around another ATmega328P MCU (IC2) with Arduino Uno bootloader, voltage regulator 7805 (IC3), 16x2 LCD character module display (LCD1), 433MHz receiver (RX1) and a few other components. The receiver checks the code word sent by the transmitter unit and displays temperature and humidity on the LCD.

If the transmitter stalls or its power supply gets interrupted, there is no way for the receiver to know whether the incoming

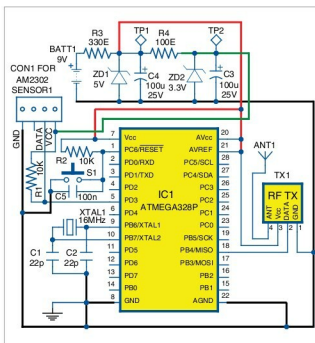


Fig. 2: Circuit diagram of the transmitter unit